

Budget-Adaptive Routing

Skipping the Weak When the Strong Answers Anyway

Wei Geng^{1,2} **Nitinder Mohan**² **Jörg Ott**¹

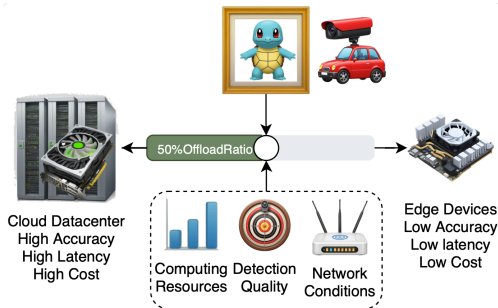
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Denver, CO, USA, August 2026

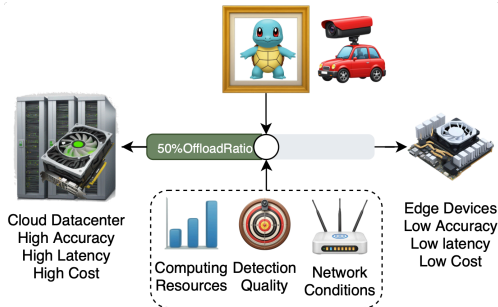
Background: Selective Obj. Det. Offloading

- **Weak** on device, **strong** in cloud.
- **Estimator** picks *weak/strong* per frame.
- Budget ρ : how many frames may offload.

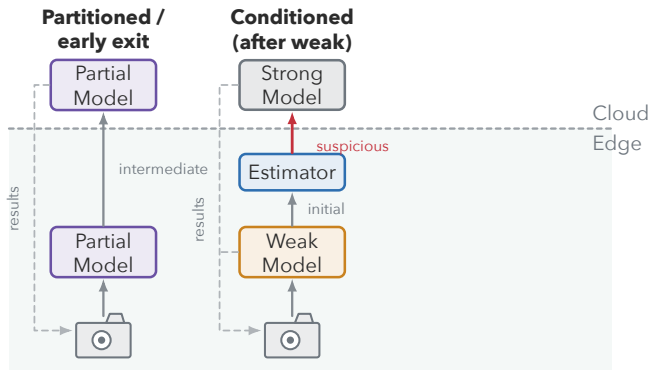


Background: Selective Obj. Det. Offloading

- **Weak** on device, **strong** in cloud.
- **Estimator** picks *weak/strong* per frame.
- Budget ρ : how many frames may offload.
- **Problem:** Design an estimator to achieve *higher accuracy* while keep *lower compute/latency*?

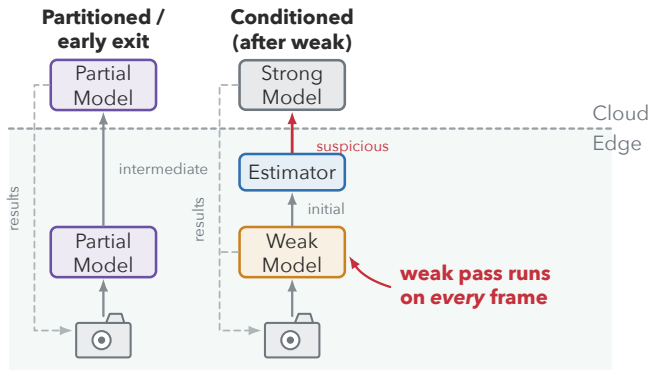


Related Work: *After or Before the Weak Pass*



- Early exits inside weak net.
- Regress on proposal features. (EdgeML [Qiu SEC'24], DCSB [Cao ICDCS'23])

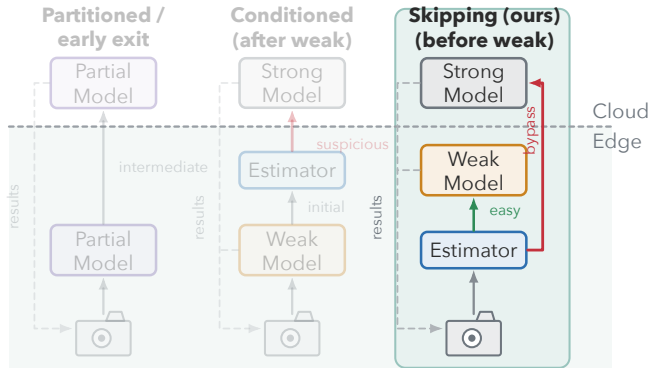
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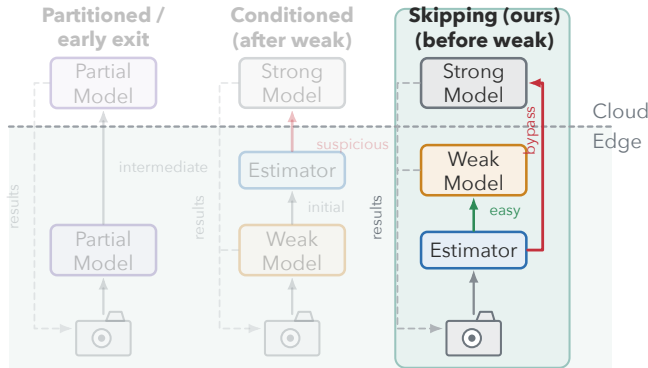
Common pattern: **weak pass on every frame** (*weak-conditioned*).

Approach: Deciding Before the Weak Pass



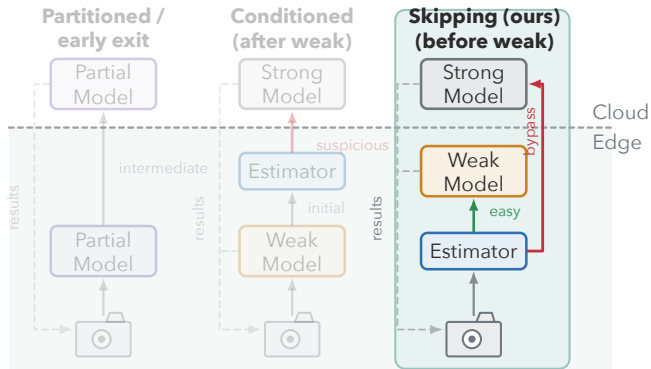
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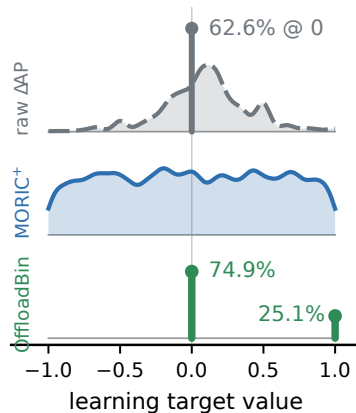
Approach: Deciding Before the Weak Pass



- A *weak-skipping* estimator reads **raw pixels** first.
- Offloaded frames *skip* the weak pass.
- Fallback path → Branchy load balancing alike

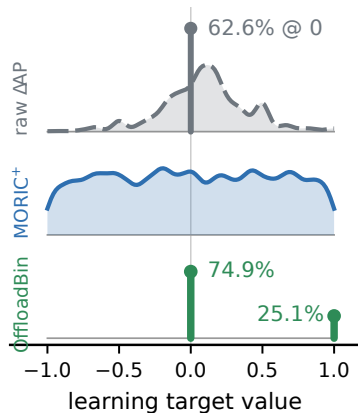
Approach: Simplified Binary Signal

- AP (Average Precision) measures how **accurately** (precision) and **completely** (recall) a detector finds objects.

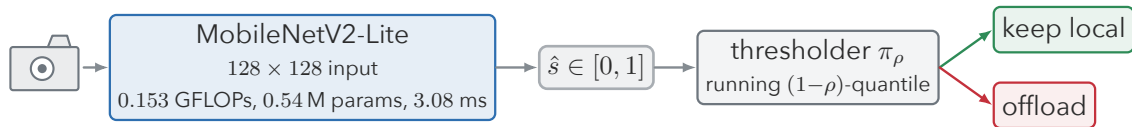


Approach: Simplified Binary Signal

- AP (Average Precision) measures how **accurately** (precision) and **completely** (recall) a detector finds objects.
- Reward: per-frame Δ AP. (delta between offload or not)
 - Concentrate
 - Flatten (MORIC+, calculate the dataset-level AP delta)
 - **Binary (OffloadBin, binarize the delta AP)**



Approach: Lightweight Estimator



- **Setup:** VOC test set, MobileV3 vs ResNet50. EdgeML is the default baseline.
- Enforced by an offload ratio ρ , estimator decides which detector detects.
- **Finally calculate the system overall performance**
 - Routing quality: mAP over the budget sweep
 - Compute usage and latency: GFLOPs and wall-clock time

Evaluation: Routing Quality

Estimator	Rank corr.	Peak mAP@0.5	Sweep AUC	Estimator GFLOPs
Always weak / strong	–	0.760 / 0.791	0.760 / 0.791	0
Oracle (Δ AP)	–	0.827	0.816	–

Spearman vs oracle; AUC = mAP- ρ area; (+4.49) = weak pass still paid.

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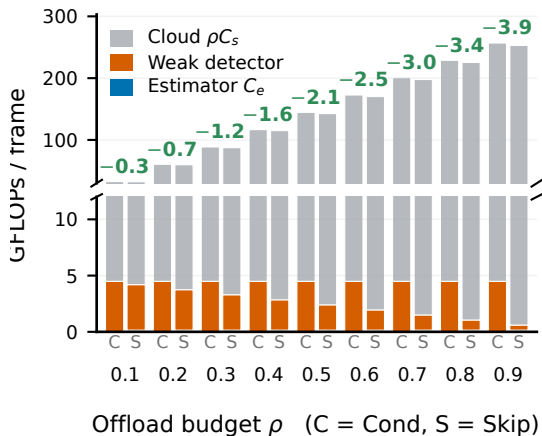
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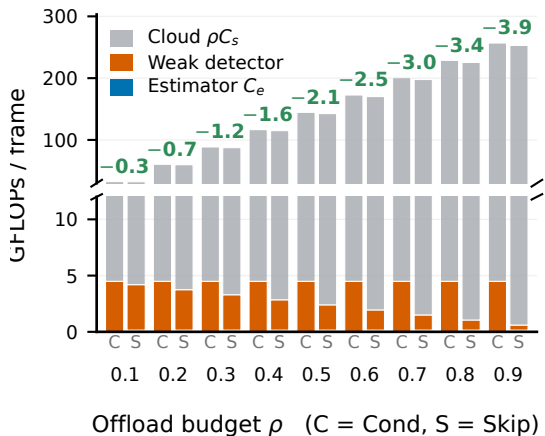
Our before-weak estimator achieves best routing quality (mAP@0.5)

Evaluation: Compute and Latency Savings

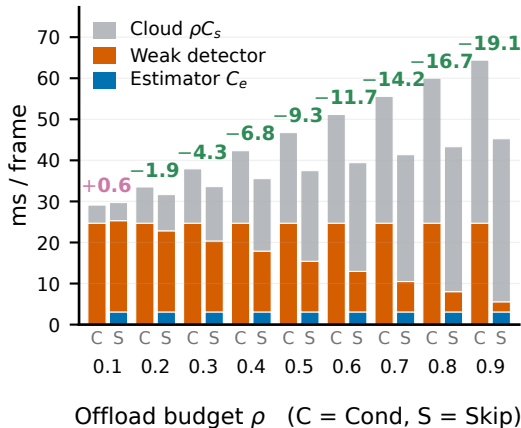


(a) Compute: up to -3.9 GFLOPs (break-even $\rho^* \approx 0.03$)

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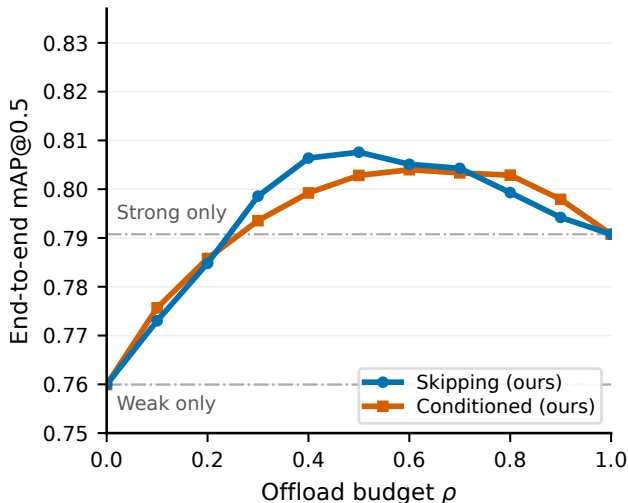
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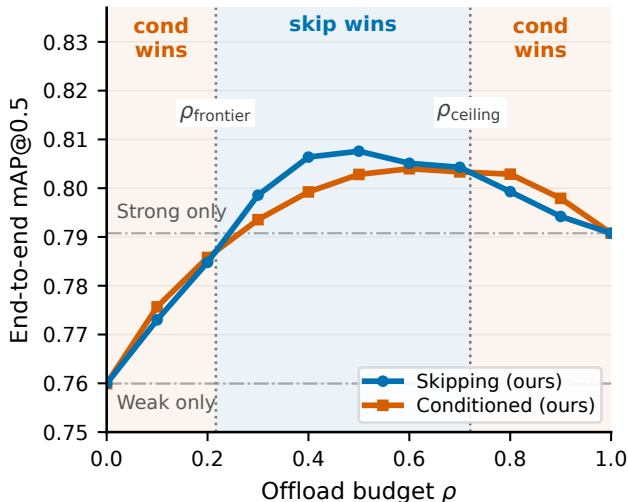
(b) Latency: up to -19.1 ms, $\sim 30\%$ of GPU time
(break-even ≈ 0.12)



Approach: But Neither Placement Dominates

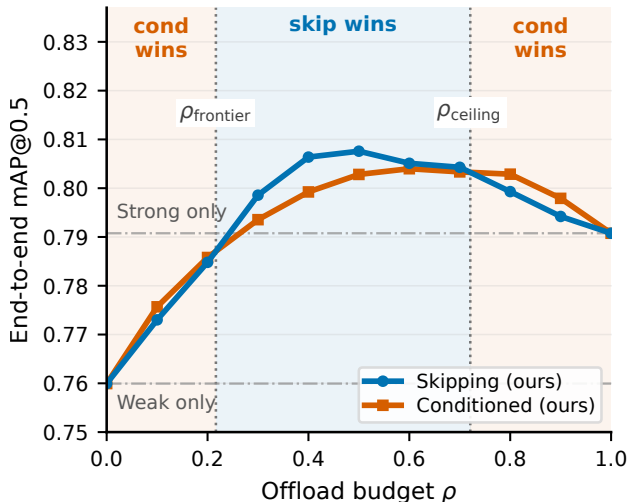


Approach: But Neither Placement Dominates



- Low ρ : *weak-conditioned* wins; features are free.
- Mid ρ : *weak-skipping* wins; less compute.
- High ρ : *weak-conditioned* wins again; few local frames matter.

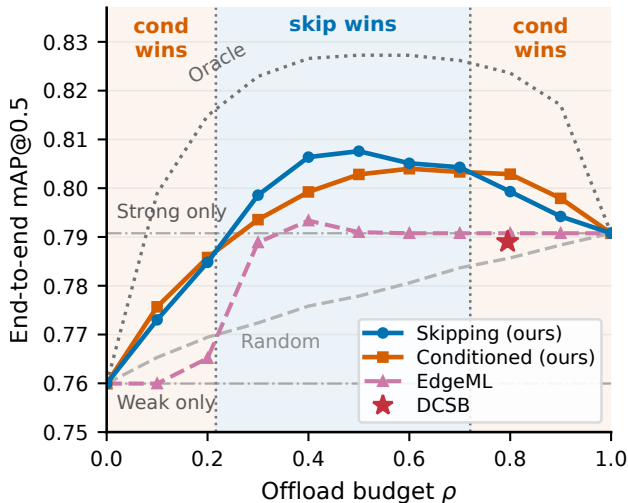
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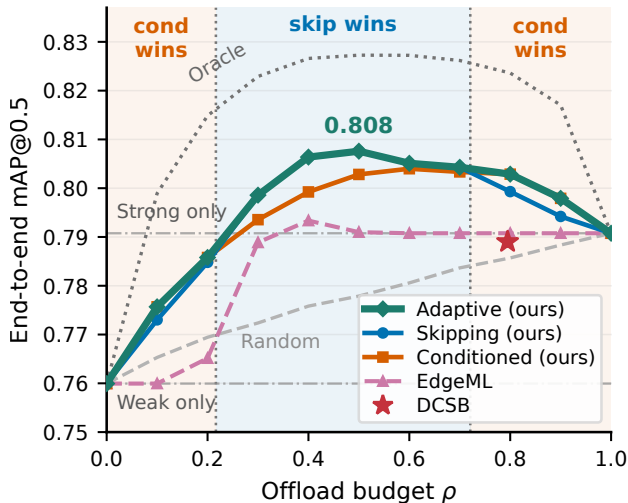
- Low ρ : *weak-conditioned* wins; features are free.
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Fixed placement is not enough.

Evaluation: *Budget-adaptive Envelope*



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■ Best across budget sweep;
 $AUC_{\rho} = 0.796$.

To Summarize

1. Placement matters: after-weak estimation pays a tax.
2. Lightweight *weak-skipping* estimator works from both routing quality and compute (latency) perspectives
3. *Budget-adaptive* estimation follows the budget and cuts latency.
4. Future works:
 - ☐ Extensive evaluations on more datasets, detectors.
 - ☐ Extend to other tasks (e.g., segmentation).
 - ☐ Consider dynamic budgets and Multi-Edge.

Authors

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Thank you!

Backup

Backup: full routing quality (VOC test, seed 42)

Estimator	ρ_s	Peak mAP	AUC $_{\rho}$	GFLOPs
Always weak	0.000	0.760	0.760	0
Always strong	0.000	0.791	0.791	0
Uniform random	0.000	–	0.777	0
Oracle (Δ AP)	–	0.827	0.816	–
EdgeML (cond)	0.138	0.793	0.784	≈ 0
DCSB (cond)	0.359	0.789	–	≈ 0
XGBoost / MORIC (cond, ours)	0.472	0.804	0.794	≈ 0
MV2-Lite + MORIC ⁺ (skip, ours)	0.350	0.801	0.790	0.15
MV2-Lite + OffloadBin (skip, ours)	0.557	0.808	0.795	0.15
Adaptive (ours)	–	0.808	0.796	0.15 [†]

[†] skipping-branch cost; the *weak-conditioned* branch additionally pays $C_w = 4.49$ GFLOPs. *Weak-conditioned* methods pay C_w upstream on every frame.

Backup: mAP@0.5 across the full budget sweep

ρ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Weak / Strong only	0.760 / 0.791 (constant)								
Random	0.765	0.769	0.772	0.776	0.778	0.781	0.784	0.786	0.788
Oracle	0.799	0.815	0.823	0.827	0.827	0.827	0.826	0.824	0.817
EdgeML	0.760	0.765	0.789	0.793	0.791	0.791	0.791	0.791	0.791
XGBoost+MORIC (ours)	0.776	0.786	0.794	0.799	0.803	0.804	0.803	0.803	0.798
MV2+MORIC ⁺ (ours)	0.771	0.778	0.785	0.791	0.796	0.800	0.801	0.801	0.798
MV2+OffloadBin (ours)	0.773	0.785	0.799	0.806	0.808	0.805	0.804	0.799	0.794
Adaptive (ours)	0.776 ^C	0.786 ^C	0.799 ^S	0.806 ^S	0.808 ^S	0.805 ^S	0.804 ^S	0.803 ^C	0.798 ^C

Superscript on the Adaptive row: selected branch (S = *weak-skipping*, C = *weak-conditioned*). Bold = column best.

Backup: learning-target ablation trunk fixed at MobileNetV2-Lite

Target / loss	Type	ρ_s	Peak mAP	ratio-err
OffloadBin (focal)	classification	0.557	0.808	0.009
TopQuartile (focal)	classification	0.389	0.800	0.007
MORIC ⁺	regression	0.350	0.801	0.018
HighIoUGain	regression	0.345	0.794	0.008
F1Gain	regression	0.306	0.791	0.006
RescueRatio	regression	0.291	0.792	0.004
... 7 more regression variants		0.169-0.313	≤ 0.801	

- Both classification targets beat every regression variant: the benefit is a sparse *partition*, not a smooth magnitude field.
- Budget tracking is uniformly tight ($\text{ratio-err} \leq 0.024$), so the gap is the target, not calibration.

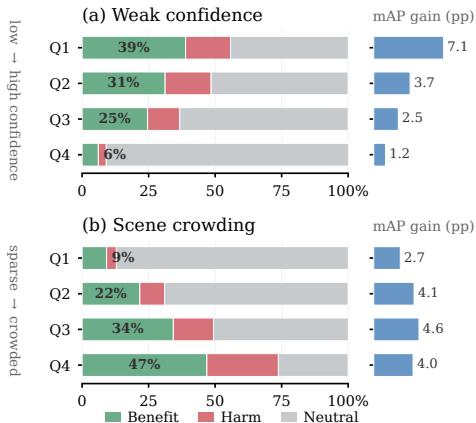
Backup: backbone ablation

learning target fixed at OffloadBin (focal)

Backbone	GFLOPs	Params (M)	ρ_s	Peak mAP
MobileNetV2-Lite	0.153	0.54	0.557	0.808
EfficientNet-B0-Lite	0.176	0.85	0.235	0.795

- The larger trunk does *not* route better – consistent with the premise that detecting difficulty needs little capacity.
- Broader sweep left to future work.

Backup: where offloading helps



- Benefit concentrates $6.4\times$ in the least-confident quartile (38.9% vs 6.1%) and $5.1\times$ in the most crowded scenes (46.9% vs 9.2%).
- Median gain is 0 in every stratum: the benefit is a sparse partition $\Rightarrow$ classify membership, don't regress magnitude.

Backup: setup and cost model

Detectors (PASCAL VOC, Nvidia A40)

	GFLOPs / ms	mAP@0.5
Weak: FRCNN-MobileNetV3	4.49 / 24.70	0.760
Strong: FRCNN-ResNet50v2	280.37 / 44.12	0.791
Estimator: MV2-Lite	0.153 / 3.08	–

Cost model

$$T_{\text{cond}} = C_w + C_e^{\text{cond}} + \rho C_s; \quad T_{\text{skip}} = C_e^{\text{skip}} + (1-\rho)C_w + \rho C_s$$
$$\text{tax} = \rho C_w - (C_e^{\text{skip}} - C_e^{\text{cond}}); \quad \text{breakeven } \rho^* = C_e^{\text{skip}} / C_w \approx 0.03$$

Evaluation

VOC test, $N = 3105$, single pass, seed 42, mAP at IoU 0.5. Oracle = per-frame ground-truth Δ AP ranking. Weak/strong gap is only 3.1 pp – routing here is *hard*; larger gaps make the benefits more pronounced.

Adaptive arbiter

$\alpha(\rho)$ piecewise constant with two offline-tuned switch points on a held-out split: $\rho_{\text{frontier}} = 0.3$, $\rho_{\text{ceiling}} = 0.8$.

Backup: taxonomy of related work

- *Weak-conditioned* estimators – EdgeML (top-25 proposal regression), DCSB (hand-crafted discriminator), confidence thresholds, learned detection embeddings. Rich signal, but the tax is unconditional.
- Mid-network gates / early exit – BranchyNet, MSDNet, SDN: estimator fused *into* the weak net; cheaper on easy frames but never skips the early pass.
- Split computing – Neurosurgeon, SPINN: partition one network across device and cloud; related but a different axis (where to cut, not whether to run).
- Cascades – Viola-Jones to Cascade R-CNN: the cheap stage *produces predictions*; our estimator only *decides where to send the frame*.
- *Weak-skipping* estimators – image-complexity heuristics existed, but were considered too weak for detection routing; our results revise that premise.